HETEROGENEOUS LLM FINE-TUNING

CENTRAL SERVER (SPA Aggregation)

1. Reconstruct
Input: Ranks
$\Delta W = B \times A$

2. SVD
Extract Signal
$W = U \Sigma V^T$

3. Project
Adaptive Rank
r=4, r=16...

RESULTS (Yelp)
- ✓ **Acc: 63.8%** (vs 61%)
- ✓ **Cost: -25%**
- ✓ **PPL: 11.94**

Client A (Edge)
8GB | r=4
Frozen LLM
LoRA (Small)
$B_4 \times A_4$

Client B (Mid)
16GB | r=8
Frozen LLM
LoRA (Med)
$B_8 \times A_8$

Client C (High)
24GB | r=16
Frozen LLM
LoRA (High)
$B_{16} \times A_{16}$

Client D (Server)
48GB | r=32
Frozen LLM
LoRA (Large)
$B_{32} \times A_{32}$

HETEROGENEOUS CLIENT LAYER (Federated)